

► Generative AI and Jobs

A Refined Global Index of Occupational Exposure

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Abstract

This study updates the ILO's 2023 Global Index of Occupational Exposure to Generative AI (GenAI), incorporating recent advances in the technology and increasing user familiarity with GenAI tools. Using a representative sample from the 29,753 tasks in the Polish occupational classification system and a survey of 1,640 people employed in each 1-digit ISCO-08 groups, we collect 52,558 data points regarding perceive potential of automation for 2,861 tasks. We then compare this input with a survey and several rounds of Delphi-style discussions among a smaller group of international experts. Based on this process, we create a repository of knowledge about task automation that goes beyond national specificities and use it to develop an AI assistant able to predict scores for tasks in the technical documentation of ISCO-08.

Our 2025 scores are presented in a revised framework of four progressively increasing exposure gradients, with a new set of global estimates of employment shares exposed to GenAI. Clerical occupations continue to have the highest exposure levels. Additionally, some strongly digitized occupations have increased exposure, highlighting the expanding abilities of GenAI regarding specialized tasks in professional and technical roles.

Globally, one in four workers are in an occupation with some GenAI exposure. 3.3% of global employment falls into the highest exposure category, albeit with significant differences between female (4.7%) and male employment (2.4%). These differences increase with countries' income (9.6% female vs 3.5% male in Gradient 4 in HICs), and so does the overall exposure (11% of total employment in LICs vs 34% in HICs). As most occupations consist of tasks that require human input, transformation of jobs is the most likely impact of GenAI. Linking our refined index with national micro data enables precise projections of such transformations, offering a foundation for social dialogue and targeted policy responses to manage the transition.

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► Introduction

In August 2023, the ILO published the first global employment estimates of potential occupational exposure to generative artificial intelligence (GenAI) (Gmyrek et al., 2023). This research was prompted by concerns – that have featured prominently in the media – of the potential for the technology to replace large shares of knowledge work.

Since the release of the 2023 estimates, there have been further technological advances in GenAI, including the introduction of Large Language Models (LLMs) with expanded capabilities in image recognition and audio and video production. Moreover, many new GenAI applications have been developed as a result of the possibility of linking GenAI models with Application Programming Interface (API) tools. Indeed, GenAI has been likened to a general-purpose technology, meaning that it may continue generating new applications and innovations across various sectors, driving long-term economic and societal transformations.

Despite this rapidly developing landscape, it is important for policy makers to develop deeper understanding of how technological advances might impact existing jobs. To address this need, the Research department of the ILO has partnered with NASK, the National Research Institute of the Ministry of Digital Affairs in Poland, to develop a more precise assessment of potential effects of GenAI on employment across countries.

This study builds on the method developed by the ILO (Gmyrek, Berg and Bescond, 2023 - GBB hereafter), which used task descriptions from the 4-digit International Standard Classification of Occupations (ISCO-08) and the GPT-4 model to estimate task-level and occupational scores of exposure to GenAI technology and then derives global employment estimates based on official ILO labour force data. But the study includes several key refinements.

The first step in our analysis is based on Poland's national 6-digit classification of occupations that includes nearly 30,000 tasks. This expands tenfold the number of tasks given in the ISCO-08 structure and enables a fine-grained assessment of tasks' automation potential, based on a more up-to-date set of occupations and tasks than those captured in the ISCO-08 system. As Poland's 6-digit classification is aligned with the 4-digit ISCO-08, the occupational scores can be compiled at the 4-digit level allowing a straightforward comparison and updating of the 2023 estimates of the ILO (GBB). In addition, the study improves the methodological framework of GBB, by drawing on the combined strengths of human and AI abilities, with an additional layer of expert verification.

Specifically, we surveyed 1,640 people currently employed in Poland in each ISCO-08 1-digit group to rank the automation potential of a representative sample of tasks that pertain to their occupational group. As the second step, we asked a mix of national and international experts to conduct a detailed review of a large sample of tasks previously assessed in the survey to validate or adjust the level of automation scores through an iterative process. Since we maintain the hierarchical link to ISCO-08 and focus on technological feasibility rather than specific country context, by combining opinions of local and international experts, we create a system that enables more precise predictions of the potential of automation of individual occupational tasks with GenAI technology, including at the international level. To achieve that, we input the human scores into an AI model, to generate scores reflecting human judgement for nearly 30,000 individual tasks pertaining to 2,500 occupations at 6-digit level in Poland. We then use the same AI predictor to re-generate scores for the tasks in ISCO-08, previously provided by GBB (2023), thereby adjusting the 2023 index of the ILO, which now benefits from both the input of people